

TUOTUOHE, China

WMO: 560040

Lat: 34.2161N

Lon: 92.4385E

Elev: 4535

StdP: 57.46

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.0	-24.8	-36.4	0.2	-17.6	-34.7	0.2	-15.7	18.5	-1.4	16.6	-4.2	1.4	270	0.890

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	17.5	6.3	15.8	6.1	14.4	5.8	8.6	14.2	8.0	13.4	7.3	12.2	4.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.4	10.6	10.8	5.8	10.1	10.1	5.2	9.7	9.6	39.6	14.3	37.7	13.5	35.7	12.5	11.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.1	12.0	10.2	-30.2	20.5	3.2	1.7	-32.5	21.7	-34.4	22.7	-36.2	23.6	-38.5	24.9
			-30.6	9.8	3.1	0.6	-32.8	10.3	-34.6	10.7	-36.3	11.0	-38.5	11.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.5	-14.5	-12.0	-7.8	-2.8	2.2	6.3	9.0	8.6	5.1	-2.0	-9.2	-13.3
	DBStd	8.90	3.72	3.47	3.33	3.10	2.49	2.56	2.15	2.19	2.57	3.75	3.26	3.68
	HDD10.0	4585	758	615	553	383	243	114	46	54	146	372	577	724
	HDD18.3	7601	1017	848	811	633	502	362	291	302	396	630	827	982
	CDD10.0	26	0	0	0	0	0	2	13	11	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.8	4.3	4.8	4.8	4.3	4.0	3.6	3.3	3.1	2.9	3.2	3.6	3.8
Precipitation	PrecAvg	266	1	1	2	5	15	51	81	64	41	10	1	1
	PrecMax	389	4	5	10	15	55	102	174	138	82	79	4	7
	PrecMin	163	0	0	0	0	0	13	14	8	7	0	0	0
	PrecStd	58	1	1	2	3	12	21	30	30	17	15	1	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.6	2.7	7.3	10.8	16.0	18.4	19.7	19.3	15.5	12.5	3.8	2.0
		MCWB	-6.3	-5.7	-3.2	-1.2	1.9	5.1	7.4	6.6	5.8	0.9	-4.8	-5.9
	2%	DB	-1.9	0.0	4.1	8.7	12.7	15.4	17.5	17.3	13.7	8.7	1.7	-1.1
		MCWB	-8.4	-7.1	-4.7	-1.9	0.9	5.4	6.9	6.6	5.1	-0.2	-5.8	-7.9
	5%	DB	-4.4	-1.9	2.2	6.8	10.6	13.6	15.9	15.6	12.2	6.3	0.1	-3.3
		MCWB	-9.9	-8.2	-5.6	-2.6	0.2	5.0	6.6	6.4	4.8	-1.2	-6.7	-9.2
	10%	DB	-6.5	-3.9	0.4	5.0	8.7	11.8	14.3	13.9	10.6	4.3	-1.7	-5.5
		MCWB	-11.1	-9.3	-6.5	-3.3	-0.2	4.4	6.3	6.2	4.1	-2.2	-7.6	-10.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.2	-5.4	-2.9	-0.2	4.2	8.5	9.6	9.2	7.8	2.7	-4.2	-5.9
		MCDB	1.3	1.9	5.9	7.3	9.8	13.8	15.5	14.9	12.9	8.3	2.4	1.5
	2%	WB	-8.3	-6.9	-4.3	-1.2	2.6	7.3	8.8	8.5	6.8	1.1	-5.5	-7.7
		MCDB	-2.4	-0.5	3.1	6.5	8.4	12.7	14.5	14.1	11.9	6.5	0.8	-1.5
	5%	WB	-9.7	-8.0	-5.3	-2.0	1.6	6.5	8.2	7.9	6.0	-0.1	-6.4	-9.1
		MCDB	-4.7	-2.4	1.5	5.3	7.3	11.5	13.6	13.2	10.8	4.5	-0.5	-3.5
	10%	WB	-11.1	-9.1	-6.3	-2.8	0.7	5.6	7.5	7.2	5.1	-1.3	-7.4	-10.4
		MCDB	-6.7	-4.0	-0.1	3.9	6.4	10.0	12.4	12.0	9.1	3.1	-2.0	-5.7
Mean Daily Temperature Range	5% DB	MDBR	15.4	15.1	15.1	13.9	11.9	10.3	10.2	10.1	10.4	11.8	15.0	15.8
		MCDBR	17.8	17.8	17.3	17.0	16.0	13.0	12.7	13.1	12.6	14.0	16.9	18.4
		MCWBR	12.7	12.2	10.4	8.5	6.5	5.0	4.5	4.6	5.3	6.1	10.6	12.9
	5% WB	MCDBR	17.4	17.0	16.2	14.4	12.2	10.2	9.5	9.6	10.1	11.8	16.1	17.9
		MCWBR	12.6	11.8	10.0	7.4	5.7	4.7	4.0	4.0	4.7	5.4	10.3	12.7
Clear-Sky Solar Irradiance	taub	0.174	0.195	0.237	0.278	0.295	0.290	0.287	0.280	0.252	0.217	0.181	0.168	
	taud	2.624	2.562	2.435	2.323	2.338	2.460	2.535	2.550	2.597	2.626	2.640	2.676	
	Ebn at Noon	1095	1089	1049	1008	987	986	988	995	1020	1046	1076	1091	
	Edh at Noon	71	85	107	127	127	112	103	100	90	79	69	63	
All-Sky Solar Radiation	RadAvg	3.36	4.12	5.22	6.09	6.65	6.50	6.56	6.04	5.37	4.63	3.87	3.31	
	RadStd	0.29	0.33	0.29	0.32	0.30	0.26	0.39	0.33	0.29	0.30	0.16	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.66	+1.63	N/A	N/A	+0.37	N/A	-244	-254	N/A	N/A
(47)	Regional (1 neighbor)	+0.56	+1.12	N/A	N/A	+0.36	+0.26	-244	-254	N/A	N/A

Nomenclature: See separate page